Climate and Disaster Risk Screening (CDRS) – Water Project

Project: Blantyre Water and Sanitation Project (SoP-1)

Sector: Water supply and sanitation

3.1. Climate Policy Context

3.1a Is climate change a principal or significant motivation for undertaking this project?

Response: Yes (Principal Objective)

- ➔ Appraiser Assessment Grade: Pass/Unclear/Fail

3.1b How will the project contribute to Malawi's climate change objectives?

Selection cascade: Adaptation → Plan and implement adaptation actions toward an increased resilience of the most vulnerable Malawians → Water → Water harvesting and storage in drought-prone areas

Explanation: The project rehabilitates and upgrades water production and transmission (e.g., Mudi intake and treatment, storage, and pressure management) and reduces losses through NRW programs, improving resilience to flood-driven turbidity events and drought-related variability for vulnerable urban communities. It also supports sanitation upgrades and methane capture, strengthening adaptation co-benefits for public health.

- ➔ Appraiser Assessment Grade: Pass/Unclear/Fail

3.2. Vulnerability Assessment

3.2a [EXPOSURE] Hazards affecting the project location

Hazard	Exposure Rating	Justification
1. Flood (river or urban)	High	Blantyre experienced >3-day supply outages after Cyclone Ana (2022) due to sediment load and damage; future intense rainfall increases flood/turbidity risk along Shire/Mudi systems.
2. Extreme Heat	Moderate	Rising temperatures increase demand and stress on pumping and storage; heatwaves exacerbate evaporation and water quality risks.
3. Wildfire	Low/No	Urban setting; limited interface with fire-prone catchments.
4. Volcano	Low/No	No known volcanic centers affecting Blantyre system.
5. Earthquake	Low/No	Seismic hazard not a dominant risk for the assets considered.
6. Landslide/Mudslide	Moderate	Hillslope intakes and transmission corridors face localized slope instability during extreme rainfall.
7. Cyclone / Severe Storm	Moderate	Tropical systems have caused turbidity spikes and infrastructure disruption (e.g., 2022 events).
8. Drought/Water Scarcity	Moderate	Demand-growth combined with seasonal deficits and drought episodes cause supply gaps; alternative sources and storage required.
9. Other (Specify)	—	—

➔ Appraiser Assessment Grade: Pass/Unclear/Fail

3.2b [SENSITIVITY] Component sensitivity to hazards

Project Component	Sensitivity Rating	Cause of vulnerability
Physical infrastructure and onsite assets	High	Pumps, control panels, and treatment units are highly sensitive to inundation and sediment spikes; 0.5–1.0 m flooding could cause full outages and equipment damage.
Inputs needed for operation	High	Grid electricity supply interruptions during storms halt pumping; chemical deliveries can be delayed by flooded roads.
Networks and linkages to external systems	Moderate	Single transmission corridors and limited storage redundancy create bottlenecks during extreme events; telecoms outages degrade SCADA.
Outputs produced by the project	Moderate	Service continuity relies on uninterrupted pumping and treatment; moderate shocks reduce hours/pressure and service quality.

➔ Appraiser Assessment Grade: Pass/Unclear/Fail

3.2c [ADAPTIVE CAPACITY] Ability to withstand upper-bound hazard severities

Project Component	Adaptive Capacity	Justification
Physical infrastructure and onsite assets	Moderate	Rehabilitations and elevation of critical equipment reduce vulnerability under moderate floods; however, upper-bound storm events can still force shutdowns until sediment subsides.
Inputs needed for operation	Moderate	One backup generator provides limited autonomy; inventories buffer short disruptions (2–3 weeks), but prolonged access/power outages constrain operations.
Networks and linkages to external systems	Low	Unpaved/at-risk access routes and single feeders limit redundancy; prolonged disruptions likely

		after severe floods/landslides.
Outputs produced by the project	Moderate	SOPs allow controlled reductions and rapid restart after moderate shocks, but extended outages may occur after severe events.

Overall Climate and Disaster Vulnerability Assessment Rating: High

➔ Appraiser Assessment Grade: Pass/Unclear/Fail

3.3. Risk Management Framework

3.3a [EXISTING ADAPTATION MEASURES]

Measure	Yes / No	Analysis
(i) Structural measures	Yes	Rehabilitation/upgrade of Mudi intake and treatment; transmission and storage upgrades; pressure management; drainage and flood risk provisions integrated in design.
Policy Development	Yes	Water safety plans and business continuity planning included under institutional strengthening.
Emergency Planning	Yes	Emergency operation procedures and contingency operations are established for extreme-weather events.
Long-Term Strategic Planning	Yes	Utility modernization and tariff framework support resilience investments.
Maintenance and Operations	Yes	Leak detection, pressure management, and NRW teams institutionalized.
Capacity Building, Training, and Outreach	Yes	Training for operations, monitoring, and customer engagement.
Data Gathering, Monitoring, and Information Systems	Yes	SCADA, GIS, and hydraulic modelling to detect leaks/pressure anomalies and manage shocks.
(iii) Other measures (specify)	Yes	Methane capture at WWTPs reduces environmental and public-health risks during floods.

➔ Appraiser Assessment Grade: Pass/Unclear/Fail

3.3b [PLANNED ADAPTATION MEASURES]

Targeted hardening of vulnerable transmission segments (e.g., elevating low points; slope stabilization), increased storage redundancy, and secondary power/feeders where feasible to reduce outage duration under upper-bound events.

➔ Appraiser Assessment Grade: Pass/Unclear/Fail

3.3c [RESIDUAL RISK MANAGEMENT]

Risk Management Strategy	Yes / No	Analysis
Avoidance	No	Core assets must remain near service areas; full relocation is not feasible.
Minimization	Yes	Contingency operations, pre-positioned spares, staged restart protocols, and mutual aid arrangements minimize service downtime.
Compensation / Risk Transfer	Yes	Insurance coverage for key equipment; emergency budget provisions for rapid repair/replacement.
Not Applicable	No	Material residual risks remain for rare, severe events.

➔ Appraiser Assessment Grade: Pass/Unclear/Fail

Project Impact on the Climate and Environment

3.4 Climate-related impacts if implemented as planned

Measure	Response	Justification
a) Net change in GHG emissions over lifecycle	Yes (reduction)	NRW reduction and 3,000 kW solar PV reduce energy-related emissions; methane capture at WWTPs reduces CH_4.
b) Significant GHG emissions from construction/energy use	Yes	Civil works and pumping consume energy; emissions occur during construction and operation.
c) Emissions of CO_2/CH_4/N_2O/FGases	Yes	Operational energy (CO_2); wastewater CH_4 without capture; reduction expected after upgrades.
d) Removal/degradation of	No	Works focus on existing

carbon sinks or land-use change		sites and corridors.
e) Changes in hydrology or natural systems increasing risk	Unlikely	Upgrades mitigate flood-related service failures; stormwater managed within design.
g) Changes in local microclimate/ecosystems	Unlikely	Localized works; sanitation improvements reduce pollution.
h) Lock-in of high-carbon infrastructure or acceleration to low-carbon	Acceleration to low-carbon	Energy-efficiency, solar PV, and process optimization accelerate low-carbon transition.

➔ Appraiser Assessment Grade: Pass/Unclear/Fail

3.5 Strategies to reduce climate-related impacts (mitigation)

Measure	Response	Justification
Are any mitigation strategies in place?	Yes	See below selections.
a) Energy-efficiency measures	Yes	Pressure management, pump efficiency upgrades, leakage control.
b) Low-carbon/renewable energy	Yes	Solar PV for priority pumping/boosters.
c) Low-emission construction/materials	Yes	Optimized earthworks and durable materials to minimize maintenance.
d) Design changes reduce emission sources	Yes	Network reconfiguration reduces head losses and energy intensity.
e) Land-use/ecosystem protection included	Yes	Avoids new clearances; protects riparian zones near intakes.
f) Operational practices to reduce emissions	Yes	Fleet management and condition-based maintenance.
g) Waste/resource-efficiency measures	Yes	Improved sludge handling; recycling at RPEL.

➔ Appraiser Assessment Grade: Pass/Unclear/Fail

3.6 Other environmental impacts and planned investigations

Construction: dust, noise, trenching, spoil, temporary drainage alterations; Operation: wastewater/sludge handling, localized odor, and solid waste. Planned ESIA/ESMP, stormwater analyses, and dam safety assessments to guide mitigation and monitoring.

➔ Appraiser Assessment Grade: Pass/Unclear/Fail

3.7 Effects on vulnerable communities

Likely impact	Response	Justification
a) Increase exposure/sensitivity or reduce adaptive capacity	No	
b) Reduce exposure or strengthen adaptive capacity	Yes	More reliable, safer water and sanitation services reduce health risks (e.g., cholera) and service interruptions during floods.
c) Unlikely to affect	No	
d) Uncertain	No	

➔ Appraiser Assessment Grade: Pass/Unclear/Fail

3.8 Climate-related indicators (SMART)

- Number of flood-related water supply interruptions per rainy season (target: ≤2; source: O&M logs).
- NRW percentage (target trajectory to ≤35%);
- kWh per m^3 billed (target: ≤4.0);
- Time to restore service after severe-weather disruptions (target: ≤48 hours);
- Volume of wastewater treated with demonstrated methane reduction (annual).

➔ Appraiser Assessment Grade: Pass/Unclear/Fail